

Chapter 112 — Nocturia — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Nocturia** = voids during the **main period of intended sleep** (patient's schedule, not the clock)
- **Aetiology: Sleep CALM** — Sleep, Comorbidities, Actions, LUT dysfunction, Medications
- **Bladder diary** = gold standard for severity; guideline standard of care
- **Nocturnal polyuria (NP)**: $NPI > 20$ –33% (age-dependent) or $NUP > 90$ mL/h
- **Global polyuria** > 40 mL/kg/24 h → always investigate systemic disease
- **Ni** = NUV/MVV ; > 1 = mismatch; strongest diary predictor of severity
- **NBCi** > 1.3 → investigate small nocturnal capacity; > 2 = severe
- **Desmopressin**: only US-approved nocturia drug — **NP + ≥ 2 voids/night**
- **Hyponatraemia**: age ≥ 65 , Na < 135 , CrCl < 60 , excess dose; mostly **first 2–3 weeks**
- **Lower urinary tract drugs**: statistically significant, **clinically modest** for nocturia

NOCTURIA—WHAT, WHEN, AND WHY?

- **ICS 2002 vs 2019**: "wake ... to void" → cause-neutral count
- **Counts**: single void/night counts; **bother not part** of definition
- **First morning void: volume** → NUV ; **void** → **daytime frequency**
- $NPI = NUV/TUV$; $NUP = NUV/\text{hours of sleep}$
- **Modified Ni** = $NUV/NMVV$ — night-only data; surrogate for Ni
- $PNV = Ni - 1$ (last volume leaves as first morning void)
- $NBCi = ANV - PNV$; 0 = voids at full capacity
- **FUSP** = bed → **first nocturnal void**
- **Convenience voids**: little urgency, **smaller volumes**; uncommon
- **Example**: $NUV 800$, $MVV 200$ → $Ni 4$, $PNV 3$; 6 voids → $NBCi 3$
- **Capacity**: normal > 300 – 400 mL; $MVV/NMVV < 150$ – 300 mL small
- **Other NP cut-offs**: $NUV > 6.4$ mL/kg; > 0.9 mL/min; $NPI > 53\%$ (not ICS)
- **Global polyuria per hour**: 90 / 117 / 167 mL/h at 54 / 70 / 100 kg

RATIONALE FOR EVALUATION AND MANAGEMENT

- **Prevalence ≥ 2 voids**: 1 in 5–6 aged 20–40; up to 3 in 5 aged > 70
- **Strongest correlate**: older age
- **BACH (landmark)**: higher in Black, Hispanic adults vs White
- **Commonest cause of adult sleep disruption**: urge-driven waking
- **Timing**: first void **within 2–3 h** → hits **slow-wave sleep** (early night)
- **Sleep**: NREM $\approx 75\%$; 4–6 cycles; slow-wave 10–25%, not recovered
- **Falls/hip fractures**: nocturia among **most strongly associated LUTS**
- **Mortality**: ≥ 2 voids → worse survival vs 0–1, esp. < 65 years
- **Clinically significant**: any nocturia prompting consultation

CAUSES OF NOCTURIA: "SLEEP CALM" AND THE BLADDER DIARY

- **Gold standard**: **diary-counted ANV**; questionnaires usually overestimate
- **Key sleep measure**: **FUSP** — prolongation = treatment success
- **Duration**: EAU Female **3 days**; EAU Male ≥ 3 days; ICS no consensus
- **3–7 days** more reliable than 24 h; "**diary fatigue**" ↓ adherence
- **Night-only diary**: cannot diagnose global polyuria; not for women (EAU)
- **FVC vs diary**: diary adds intake, pads, leakage

- **Commonest mechanism**: **NP** (most adults); global polyuria uncommon
- **Krimpen (landmark)**: $NPI > 33\%$ in **92%** with nocturia, **70%** without

EVALUATION AND MANAGEMENT

- **Basic tests**: **metabolic panel, urinalysis, PVR**; PSA only if changes management
- **UTI on urinalysis**: treat → **repeat urinalysis**
- **First-line**: **optimise all Sleep CALM factors**, together
- **STOP-BANG**: ≥ 3 yes = high OSA risk; BMI > 35 , age > 50 , neck > 40 cm
- **OSA**: hypoxia → ↑ right atrial pressure → **ANP** → natriuresis
- **CPAP**: up to **≈ 2 fewer voids** — among largest effects
- **OSA history (women)**: **sleep-clinic referral** for CPAP (EAU strong)
- **Non-bothersome nocturia** still merits evaluation (EAU)
- **Non-dipping hypertension**: normal dip 10–20%; 40–50% of hypertensives
- **Non-dipping mechanism**: **aldosterone** → **nocturnal natriuresis**
- **HCTZ vs loop**: more antihypertensive, less diuretic
- **Heart failure**: third-spaced leg fluid **mobilised supine** → ↑ NUV
- **AVP**: supraoptic/paraventricular → posterior pituitary → **V2, collecting duct**
- **Polyuria–polydipsia**: central DI, nephrogenic DI, primary polydipsia
- **Diagnosis**: **arginine-stimulated copeptin** $>$ water deprivation test
- **Melatonin**: consider in **older adults**; evidence limited
- **Evening fluid**: restrict 3–4 h before bed; water diuresis onset ≈ 1 h
- **Caffeine/alcohol**: diuresis within 2–3 h; caffeine half-life 2.5–4.5 h
- **Protein meal**: 2–3 h hyperfiltration
- **Other conservative**: ↓ **sodium**, exercise/weight loss, compression stockings
- **Isolated nocturia is not a LUTS** — LUT drugs not expected to work
- **OAB trials**: nocturia in **up to 85%**
- **VA Cooperative**: only **terazosin** beat placebo (2.5 → 1.8)
- **MTOPS**: **doxazosin, combination** beat placebo; finasteride not
- **CombAT**: combination -0.5 vs tamsulosin -0.3 , dutasteride -0.4
- **REDUCE**: dutasteride $-0.15/-0.24$ voids; no sleep benefit
- **5ARI**: only if **prostate > 40 mL** (EAU)
- **PDE5i**: **do not offer** for nocturia (EAU)
- **Surgery**: TURP slightly $>$ drugs; **no new modality beats TURP**

- **Operate if:** BPH-LUTS refractory to medical therapy; counsel on nocturia
- **Antimuscarinics:** best for **severe urgency, small night volumes**
- **Only nocturnal-urgency primary trial:** fesoterodine — positive
- **Tolterodine ER nocturia trial:** primary **negative**; urge-related voids fell
- **Behavioural ≥ drugs** for nocturia; step-up trial → **behavioural first-line**
- **Third-line:** onabotulinumtoxinA, PTNS > placebo; PTNS ≈ tolterodine
- **EAU grades:** male LUTD drugs all **weak**; female antimuscarinic, PFMT **strong**
- **β3-agonists:** **no trial** with nocturia as primary outcome
- **OAB first-line:** **behavioural therapy** before drugs
- **GSM:** vaginal oestrogen; ospemifene
- **Sildenafil:** 4 mg if CrCl 30–50; **contraindicated <30**
- **Antimuscarinic CI:** urinary/gastric retention, narrow-angle glaucoma
- **Beers list:** antimuscarinics ≥65 years; mirabegron → hypertension
- **5ARI:** PSA halves in 3–6 months; **pregnancy category X**
- **Drug causes:** CCBs (oedema), SGLT-2 (glycosuria), SSRIs, lithium, ketamine

ADDITIONAL TREATMENT CONSIDERATIONS

- **Refractory:** Sleep CALM → conservative → **drugs** (selected)
- **Target:** **nocturnal volume** — helps even without NP
- **DDAVP:** **selective V2 agonist**, no pressor effect; **>80 countries**
- **DANS (FDA March 2017):** **1.66 mcg <65; 0.83 mcg ≥65/at risk**; 30 min before bed
- **ODT (FDA June 2018):** **women 25, men 50 mcg**; sublingual, no water, 60 min
- **Sodium (FDA):** baseline → ≤7 days → ≈30 days → periodic
- **Sodium (EAU):** baseline → day 3 → day 7 → 1 month → periodic
- **DDAVP contraindications:** **low baseline sodium** (EAU avoid); **loop diuretics** (US label)
- **Old high-dose tablets:** Na <130 in ≈5%
- **EAU Male:** desmopressin <65; **low dose >65** with ≥2 voids from NP

- **Furosemide:** diuresis **6–8 h**; give >6 h before bed
- **Timed diuretic:** morning/early afternoon; **level 1b** for NP

ANTIDIURETIC AND DIURETIC THERAPY: KEY TRIALS

- **Old tablets:** 50% response **34% vs 3%** (men), **46% vs 7%** (women)
- **DANS pooled (n 1333, ≥50 y):** -1.4/-1.5 vs -1.2; Na event 0/1.1%
- **ODT dose-ranging:** **women respond at lower doses**
- **ODT women 25 mcg:** -1.46 vs -1.24; Na <130 **2.2%**
- **ODT men 50/75 mcg:** -1.25/-1.29 vs -0.88; Na <130 **1.9% vs 7.4%**
- **Japan women 25 mcg:** **not significant** (only negative primary)
- **Responder cut-off:** **50%** (tablets, DANS) vs **33%** (ODT)
- **Men 75 mcg:** little extra efficacy, **more hyponatraemia** → 50 mcg
- **Proposed monitoring:** **stop if Na <135** → prevented all <130
- **Why women lower dose:** **V2 gene on X chromosome**

CLASSIC TRAPS

- Trap: first morning void is a nocturia episode — Actually: **daytime void; volume counts in NUV**
- Trap: one void/night is not nocturia — Actually: **counts; bother drives care**
- Trap: NPi >33% diagnoses NP reliably — Actually: **poor discrimination (Krimpen)**
- Trap: night-only diary is complete — Actually: **misses global polyuria**
- Trap: small 24-h capacity raises NBCi — Actually: **NBCi may be normal**
- Trap: nocturia is a LUTS — Actually: **only when LUTD contributes**
- Trap: significant P = useful nocturia effect — Actually: **fractions of a void**
- Trap: diuretics cause nocturia — Actually: **timed early, they treat NP**
- Trap: afternoon furosemide + desmopressin — Actually: **loop diuretic contraindicates DDAVP (US)**
- Trap: same desmopressin dose both sexes — Actually: **women lower; hyponatraemia at higher**